# Midterm 2

Let $f, g$ be continuous on $[a,b]$ and different. on $(a,b)$. Show that there exists at least one point $c \in (a,b)$ s.t.

$$[f(b)-f(a)]\, g'(c) = [g(b)-g(a)]\, f'(c).$$

Hint: consider $h(x) = \alpha f(x) + \beta g(x)$ for suitable $\alpha, \beta$.

Solution:

Let $h(x) = \alpha f(x) + \beta g(x)$, for $x \in [a,b]$, $\alpha, \beta \in \mathbb{R}$.

Then $h$ is continuous on $[a,b]$ because it is a linear combination of continuous functions (Theorem 2.35). [1]

$h$ is differentiable on $(a,b)$ because it is a linear combination of differentiable functions (Theorem 3.8). [1]

Therefore, we can apply the MVT to $h$ on $[a,b]$.

MVT: Let $h$ be continuous on $[a,b]$ and differentiable on $(a,b)$. Then $\exists c \in (a,b)$ s.t. [1]

$$h(b)-h(a) = h'(c)(b-a).$$

Therefore, $\exists\, c \in (a,b)$ s.t. [1]

$$h(b)-h(a) = h'(c)(b-a).$$

We calculate:

- $h(b)-h(a) = \alpha f(b)+\beta g(b)-\alpha f(a)-\beta g(a)$
  $= \alpha(f(b)-f(a))+\beta(g(b)-g(a))$. [1]
- $h'(x) = \alpha f'(x)+\beta g'(x)$. [1]

Hence,

$$\alpha(f(b)-f(a))+\beta(g(b)-g(a)) \qquad (*)$$
$$= (b-a)[\alpha f'(c)+\beta g'(c)]. \quad [1]$$

In order to obtain the equality that we want, we choose

$$\alpha = g(b)-g(a)$$
$$\beta = -(f(b)-f(a)) \quad [1]$$

so that (*) becomes

$$[g(b)-g(a)][f(b)-f(a)] - [f(b)-f(a)][g(b)-g(a)]$$

$$= (b-a)\big[(g(b)-g(a))f'(c) - (f(b)-f(a))g'(c)\big]$$

$$\Leftrightarrow \quad 0 = (b-a)\big[(g(b)-g(a))f'(c) - (f(b)-f(a))g'(c)\big]. \qquad [1]$$

Since $a < b$, we must have that

$$[g(b)-g(a)]f'(c) - [f(b)-f(a)]g'(c) = 0. \qquad [1]$$